19.8.26

Chap-6

Least Squares

no of eqns > no of variables → overdetermined system

no of eqns < no of variables → underdetermined system

Ex:

$$x_1 + 2x_2 + 4x_3 = 3$$
$$x_1 + 3x_2 + 9x_3 = 5$$
$$x_1 + 5x_2 + 25x_3 = 12$$
$$x_1 + 6x_2 + 36x_3 = 15$$

$$\underbrace{\begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix}}_{A\ 4\times 3} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 3 \\ 5 \\ 12 \\ 15 \end{bmatrix}}_{b}$$

$m \neq n$

$$\underset{m\times n}{A}\ \underset{n\times 1}{x} = \underset{m\times 1}{b}$$

$$\Rightarrow \underset{n\times n}{(A^TA)}\ \underset{n\times 1}{x} = \underset{n\times m}{A^T}\ \underset{m\times 1}{b}$$

$$\Rightarrow \underset{(n\times n)}{B}\ \underset{n\times 1}{x} = \underset{(n\times 1)}{B'} \qquad \therefore x = B^{-1}b' = (A^TA)^{-1}(A^Tb)$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 6 \\ 4 & 9 & 25 & 36 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 6 \\ 4 & 9 & 25 & 36 \end{bmatrix} \times \begin{bmatrix} 3 \\ 5 \\ 12 \\ 15 \end{bmatrix}$$

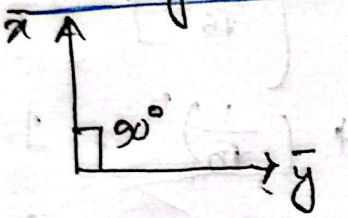
A^T 3×4 $\quad A$ 4×3 $\quad x$ $\qquad A^T$ $(3\times 4$

$$\begin{bmatrix} 4 & 16 & 74 \\ 16 & 74 & 376 \\ 74 & 376 & 2018 \end{bmatrix}_{3\times3} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 35 \\ 171 \\ 879 \end{bmatrix}_{3\times1}$$

$x_1 = -28.4 \quad x_2 = -17.77 \quad x_3 = -1.83$

Vector:

Orthogonal vector:



$\vec{x} \cdot \vec{y} = 0$

$\vec{x} = \frac{2}{\sqrt{5}}\hat{i} + \frac{1}{\sqrt{5}}\hat{j} \qquad \vec{y} = \frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}$

$\vec{x} \cdot \vec{y} = \left(\frac{2}{\sqrt{5}}\hat{i} + \frac{1}{\sqrt{5}}\hat{j}\right) \cdot \left(\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}\right)$

$= \frac{2}{5} - \frac{2}{5} = 0$

$$x = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}_{2\times1} \qquad y = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}_{2\times1}$$

$$x^T y = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}_{1\times2} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}_{2\times1}$$

$= 2/\sqrt{5} \times \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}} \times -\frac{2}{\sqrt{5}} = 0$

$$\boxed{x^T y = 0}$$

Normal vector/unit vector:

$$\vec{x} = \frac{2}{\sqrt{5}}\hat{i} + \frac{1}{\sqrt{5}}\hat{j}$$

$$|\vec{x}| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = \sqrt{\frac{4}{5} + \frac{1}{5}} = \sqrt{\frac{5}{5}} = 1$$

$$x = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \qquad x^T x = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$= \left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2 = 1$$

$$\boxed{x^T x = 1}$$

Orthonormal vectors:

Set of vectors -

i) Orthogonal to each other

ii) Normal

$$\{x, y\} = \left\{ \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix} \right\}$$

$x^T y = 0$, $x^T x = 1$, $y^T y = 1$

$\{x, y, z\}$

$x^T y = 0$ $\quad y^T z = 0$ $\quad z^T x = 0$

$x^T x = 1$ $\quad y^T y = 1$ $\quad z^T z = 1$

{w, x, y, z}

$w^T x = 0$ $\quad w^T y = 0$ $\quad w^T z = 0$

$x^T y = 0$ $\quad x^T z = 0$ $\quad y^T z = 0$

$w^T w = 1$ $\quad x^T x = 1$ $\quad y^T y = 1$ $\quad z^T z = 1$

29.8.26

Unit-I

Q-R Decomposition

$A_{m \times n} x = b$

Q R $x = b$

Q → Orthonormal matrix

R → U.T.M

$A = QR$

$\Rightarrow Q^T A = Q^T Q R$

$\Rightarrow IR = Q^T A$

$\therefore \boxed{R = Q^T A}$

$Q_{m \times n} R x = b$

$\Rightarrow Q^T Q R x = Q^T b$

$\Rightarrow I R x = Q^T b$

$\Rightarrow \boxed{Rx = Q^T b}$

$Q^T Q = I$

$$\begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix}$$

$a_0 + a_1 + a_2 = 2$

$a_0 + 2a_1 + 4a_2 = 3$

$a_0 + 3a_1 + 9a_2 = 6$

$a_0 + 4a_1 + 16a_2 = 4$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

A $\quad$ x $\quad$ b

$$\begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix}$$

Gram-Schmidt process

$A \rightarrow Q$

i) $u_1 \rightarrow q_1$

$$p_1 = u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$q_1 = \frac{p_1}{|p_1|} = \frac{1}{\sqrt{1^2+1^2+1^2+1^2}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

ii) $u_2 \rightarrow q_2$

$$p_2 = u_2 - (u_2^T q_1) q_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \left(\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} \right) \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - 5 \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 2.5 \\ 2.5 \\ 2.5 \\ 2.5 \end{bmatrix}$$

$$= \begin{bmatrix} -1.5 \\ -0.5 \\ .5 \\ 1.5 \end{bmatrix}$$

$$q_2 = \frac{p_2}{|p_2|} = \frac{1}{\sqrt{-1.5^2 + (-0.5)^2 + 0.5^2 + 1.5^2}} \begin{bmatrix} -1.5 \\ -0.5 \\ .5 \\ 1.5 \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} -1.5 \\ -0.5 \\ .5 \\ 1.5 \end{bmatrix}$$

iii) $u_3 \rightarrow q_3$

$$p_3 = u_3 - (u_3^T q_2) q_2 - (u_3^T q_1) q_1$$

$$= \begin{bmatrix} 1 \\ 4 \\ 9 \\ 16 \end{bmatrix} - \left(\begin{bmatrix} 1 & 4 & 9 & 16 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} -1.5 \\ -0.5 \\ .5 \\ 1.5 \end{bmatrix} \right) \frac{1}{\sqrt{5}} \begin{bmatrix} -1.5 \\ -0.5 \\ .5 \\ 1.5 \end{bmatrix}$$

$$- \left(\begin{bmatrix} 1 & 4 & 9 & 16 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} \right) \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

$$q_3 = \frac{p_3}{|p_3|} = \frac{1}{\sqrt{1^2 + (-1)^2 + (-1)^2 + 1^2}} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0.5 & -1.5/\sqrt{5} & 0.5 \\ 0.5 & -0.5/\sqrt{5} & -0.5 \\ 0.5 & 0.5/\sqrt{5} & -0.5 \\ .5 & 1.5/\sqrt{5} & 0.5 \end{bmatrix}$$

$R = Q^T A$

$$R = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_1 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{bmatrix}$$

$$u_1^T q_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} = 2$$

$$R = \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix}$$

$$Rx = Q^T b$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \underset{3\times 4}{\begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & -0.5/\sqrt{5} & 0.5/\sqrt{5} & 1.5/\sqrt{5} \\ 0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix}} \underset{4\times 1}{\begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}}$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 15/2 \\ \frac{9\sqrt{5}}{10} \\ -\frac{3}{2} \end{bmatrix}$$

$$2a_2 = -\frac{3}{2}$$

$$a_2 = -\frac{3}{4} = -0.75$$

$$\sqrt{5}a_1 + 5\sqrt{5}a_2 = \frac{9\sqrt{5}}{10}$$

$$a_1 = -4.65$$

$$2a_0 + 5a_1 + 15a_2 = \frac{15}{2} \quad \therefore a_0 = -2.25$$

Ans

$Q^T Q = I$

$$u = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \qquad v = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{bmatrix}$$

$$Q^T Q = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \end{bmatrix}$$

(q_1^T = first row, q_1 = first column; q_2^T = second row, q_2 = second column)

$$= \begin{bmatrix} \frac{2}{\sqrt{5}} \times \frac{2}{\sqrt{5}} + \frac{1}{\sqrt{5}} \times \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}} \times \left(-\frac{2}{\sqrt{5}}\right) \\ \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} - \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} & \frac{1}{\sqrt{5}} \times \frac{1}{\sqrt{5}} + \left(\frac{2}{\sqrt{5}} \times \frac{-2}{\sqrt{5}}\right) \end{bmatrix}$$

($q_1^T q_1$, $q_1^T q_2$, $q_2^T q_1$, $q_2^T q_2$)

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Img not A

Proof:

$$Q = \begin{bmatrix} | & | & | & \\ q_1 & q_2 & q_3 & \cdots \\ | & | & | & \end{bmatrix}$$

$$Q^T Q = \begin{bmatrix} — & q_1^T & — \\ — & q_2^T & — \\ — & q_3^T & — \\ & \cdots & \end{bmatrix} \times \begin{bmatrix} | & | & | & \\ q_1 & q_2 & q_3 & \cdots \\ | & | & | & \end{bmatrix}$$

$$= \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & q_1^T q_3 & \cdots \\ q_2^T q_1 & q_2^T q_2 & q_2^T q_3 & \cdots \\ q_3^T q_1 & q_3^T q_2 & q_3^T q_3 & \cdots \\ \vdots & \vdots & & \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & & & \ddots \end{bmatrix}$$

$$= I$$

$$q_i^T q_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Imp Proof: $A = QR$

Gram – schmidt process:

$$P_1 = u_1$$

$$P_2 = u_2 - (u_2^T q_1) q_1$$

$$P_3 = u_3 - (u_3^T q_2) q_2 - (u_3^T q_1) q_1$$

$$P_k = u_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$q_k = \frac{P_k}{|P_k|} \Rightarrow P_k = |P_k| q_k$$

$$\Rightarrow |P_k| q_k = u_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$\Rightarrow u_k = |P_k| q_k + \sum_{i=1}^{k-1} (u_k^T q_i) q_i \quad \text{---(i)}$$

Multiply (i) by q_k^T,

$$q_k^T q_k = |p_k| \underbrace{q_k^T q_k}_{1} + \sum_{i=1}^{k-1} (u_k^T q_i) \underbrace{q_k^T q_i}_{0}$$

$$\Rightarrow |p_k| = q_k^T u_k = u_k^T q_k$$

$$(i) \rightarrow u_k = (u_k^T q_k) q_k + \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$u_k = \sum_{i=1}^{k} (u_k^T q_i) q_i$$

$$u_2 = (u_2^T q_1) q_1 + (u_2^T q_2) q_2$$

$$u_1 = (u_1^T q_1) q_1$$

$$u_3 = (u_3^T q_1) q_1 + (u_3^T q_2) q_2 + (u_3^T q_3) q_3$$

$$A = \begin{bmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} (u_1^T q_1) q_1 & (u_2^T q_1) q_1 + (u_2^T q_2) q_2 & (u_3^T q_1) q_1 + (u_3^T q_2) q_2 + (u_3^T q_3) q_3 \\ \cdot & | & \end{bmatrix}$$

$$= \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix} \times \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_2 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{bmatrix}$$

$$A = Q \times R$$